

Nisong Monyimba

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github.com/NisongMonyimba/AppliedMathThesis

Education

Biomedical Engineering, M.Sc. — Arizona State University

Graduated May 2022

Ira A. Fulton Schools of Engineering; MasterCard Foundation Scholar (Accelerated Master's, 3+1+1); Dean's List: Fall 2020, Spring 2021

Biomedical Engineering, B.Sc. — KNUST

Graduated May 2021

Kwame Nkrumah University of Science & Technology; MasterCard Foundation Scholar; Provost's List; First Class Honours

Relevant quantitative coursework: Algebra, Calculus with Analysis, Calculus with Several Variables, Differential Equations, Numerical Methods, Probability and Statistics, Finite Element Modeling, C Programming, Object-Oriented Programming, Signals and Systems, Biosignal Processes and Analysis, Embedded Systems.

Since 2025, self-directed graduate-level study (independent of formal coursework) in real analysis, stochastic processes, and stochastic control theory, undertaken in connection with an independent research program in mean-field games with common noise (see Research Statement and supplementary repository).

Research Preprints

Quantitative Contraction Estimates for Common-Noise Mean-Field Games

2026

- Establishes a local contraction estimate, on an explicit time horizon, for the fixed-point map characterising equilibria of mean-field games with common noise; extended to arbitrary horizons via a standard continuation argument.
- PDF | Source

Numerical Approximation of Common-Noise Mean-Field Games via Particle Methods and Finite Differences

2026

- Combined Euler–Maruyama, particle-method, and finite-difference scheme for the coupled system characterising common-noise mean-field game equilibria; every reported convergence rate is the direct, reproducible output of executed code.
- PDF | Source & reproducible code

MSc Thesis: Quantitative Convergence Analysis of Stochastic Maximum Principles for Mean-Field Games with Common Noise

2026

- PDF | Full repository (code, tests, Lean 4 formal verification)

Master's Applied Project & Undergraduate Capstone Project

Master's Applied Project: Trophoblast-on-Chip Microfluidic Device

2022

- Investigated trophoblast growth to quantify proliferation dynamics (cell counts over time, growth curves, proliferation index) using Fiji to extract quantitative growth metrics from laboratory cell images.
- Performed statistical analysis including growth-curve comparison across conditions (two-way ANOVA), doubling-time comparison between treatments (unpaired t-test), and proliferation-marker quantification (Kruskal–Wallis).
- Developed microfluidic channel designs in SolidWorks, simulated laminar flow and shear profiles in COMSOL Multiphysics, and validated models against analytical equations.

Capstone Project: Novel ECMO Cannula Design, Arizona State University

2021

- Designed and simulated ECMO cannula flow in COMSOL Multiphysics, optimizing shear stress and velocity distribution; built a 3D model in SolidWorks and simulated flow in ANSYS Fluent.
- Quantified flow uniformity and reduced clot-formation risk by optimizing inlet geometry and flow-rate parameters.
- Performed finite element analyses (FEA) in ANSYS and SolidWorks to assess performance and safety margins under physiological load conditions.

Research & Lab Work

Research Assistant, Weaver Lab, Arizona State University

2021–2022

- Analyzed trophoblast proliferation dynamics by quantifying cell counts over time and fitting growth curves.
- Used Fiji to extract quantitative metrics from laboratory images, including colony expansion area and fluorescence intensity, supporting rigorous modeling of cellular growth patterns.
- Quantified cell proliferation rates from fluorescence microscopy data using Fiji and GraphPad Prism; developed statistical plots and regression analyses to characterize proliferation and viability.

- Developed Python-based probabilistic simulations and Monte Carlo analyses to evaluate variability in material properties and stochastic loading, improving predictive accuracy.

Research Assistant, Dr. Pizziconi's Lab, Arizona State University

2022–2023

- Designed and optimized bone fixation screws using Halpin–Tsai micromechanics models to predict composite behavior under mechanical load.
- Created and refined 3D CAD models in SolidWorks, ensuring dimensional accuracy, manufacturability, and integration with experimental validation workflows.
- Trained and mentored lab members in modeling, simulation, and experimental validation, connecting theoretical mathematics with engineering practice.
- Collaborated with interdisciplinary teams to iterate designs from simulation results; documented and presented research outcomes supporting publication efforts and potential clinical translation.

Research Assistant, Medical Device Regulation (MDR) Project, KNUST

2020–2021

- Analyzed regulatory frameworks governing medical devices across 20 African countries, identifying disparities in approval processes, safety standards, and market accessibility.
- Used publicly available regulatory data to categorize countries by regulatory maturity, from fully established authorities to emerging or informal frameworks.

Internship Experience

REU Internship, Case Western Reserve University

2020

- Studied biological design and product development principles; designed Arduino-based air-quality sensors and prototyped PCB systems using EAGLE CAD.
- Collaborated with interdisciplinary teams to apply quantitative modeling to life-science problems.

BME Internship, Komfo Anokye Teaching Hospital, Kumasi, Ghana

2019

- Supported medical equipment maintenance and calibration; performed voltage and current testing on imaging systems.
- Documented device performance data and proposed engineering solutions to minimize error variance; presented findings to senior biomedical engineers and clinicians.

Engineering Internship, Vodafone (Telecom) Headquarters, Accra, Ghana

- Designed and proposed a telemedicine smartphone application concept for patient monitoring.
- Assisted in vendor management and quality audits for customer-facing telecom hardware systems.

Professional Experience

Senior Manufacturing Engineer, Abbott Laboratories

2024–2025

- Optimized medical device grinding precision by 31.7% through DOE, ANOVA, and SPC-driven process improvements.
- Implemented Python-based control dashboards for real-time statistical quality monitoring.
- Applied Monte Carlo simulations, PCA, and variance component analysis to identify and mitigate process variability.
- Authored and executed FDA- and ISO 13485-compliant validation protocols.
- Experienced with DMAIC, CAPA, QMS, Lean Six Sigma, Pareto charts, SPC, RCA, DOE, FMEA, control charts, Gage R&R, DHF, SOP, DMF, PDR, and QSR.

Quality Engineer, Vastek Inc., San Diego, CA

2023–2024

- Designed devices and implemented engineering design practices targeting efficiency improvements; worked with a cross-functional team of medical professionals and engineers.
- Performed design analyses, requirements reviews, and revisions; wrote technical reports and protocols.
- Conducted FMEA, Bayesian risk analysis, and fault-tree analysis to quantify cascading failure probabilities.

Med Tech R&D Engineer, Mayo Clinic & Arizona State University

2022–2023

- Designed bone fixation screws using Halpin–Tsai equations and Mori–Tanaka models.
- Ran nonlinear FEA simulations to analyze fatigue, stress distribution, and deformation behavior.

Teaching Experience

Teaching Assistant, KNUST

2019

- Tutored biomedical engineering undergraduates in calculus, electrical circuits, and biostatistics; assisted with lab supervision, grading, and practical demonstrations.

Science & Math Teacher, NSMQ Team, Kwanyako Senior High School

2016–2018

- Prepared and led students for regional and national science and math competitions, emphasizing teamwork, critical thinking, and problem-solving.

Math Teacher, Witness Academy

2016–2017

- Taught core mathematics and physics; prepared junior-high students for final exams (BECE), improving average performance.

Volunteering, Community Service & Projects

- Group Homes Volunteer (Autism Support): assisted residents with learning and recreational activities.
- MasterCard Foundation Community Projects: led sanitation, clean-water, and youth education projects in Ghana and Arizona.
- Oforikrom Outreach Project: organized public school clean-ups, donations, and science tutoring.
- Vocational Training: trained four youths in electrical wiring and biofill installation.
- Personal Tutoring Initiative: mentored underprivileged students in mathematics; three students gained high-school admission and two gained college admission, all first-generation in their families.
- Organized community cleaning (gutters, weeds, trash) in Ashaiman-Zenu to reduce contamination and malaria risk.

Awards & Competitions

- MasterCard Foundation Scholarship, Arizona State University and KNUST
- MORE Fellowship Award, Arizona State University
- Vodafone Telecommunications Design Competition, 1st Place
- Best Student, WASSCE (2016), Kwanyako Senior High School
- Best Student, BECE (2013), Zenu T.M.A No. 3 JHS
- 2nd Place, KNUST Engineering & Medical School Innovation Challenge
- Dean's List, Arizona State University

Leadership & Mentorship

- Assistant Senior Prefect, Zenu T.M.A Junior High School
- Leader, NSMQ Science & Math Quiz Team, Kwanyako SHS
- Peer Mentor, MasterCard Foundation Scholars (KNUST & ASU)
- Community Project Facilitator, MasterCard Foundation Ghana
- Member, Young African Leaders Initiative (YALI)
- Club Executive, Biomedical Engineering Society (KNUST & ASU)

Certifications

- Lean Six Sigma (Green Belt)
- Design of Experiments (Minitab & JMP)
- Global Leadership & Project Management (YALI / ASU Global Leadership Academy)

Professional Memberships

Society of Manufacturing Engineers (SME) • National Society of Black Engineers (NSBE) • Biomedical Engineering Society (BMES) • ASU Global Leadership Academy • Toastmasters International • KNUST Peer Counsellors & MasterCard Peer Mentors • Ghana Engineering Students Association • TraTech

Technical Skills

Programming & Computation: Python (NumPy, SciPy, statsmodels, pandas), MATLAB, R, Git/GitHub, SQL, Java, C, LaTeX (Overleaf).

Statistical & Numerical Modeling: Monte Carlo simulation, ANOVA, DOE, SPC, PCA, GraphPad Prism, Minitab, JMP, Octave/GNU.

Engineering Simulation & CAD: SolidWorks, ANSYS Fluent, COMSOL Multiphysics, Autodesk Fusion 360, SimScale.

Other: Fiji/ImageJ, BioRender, Figma, Inkscape, LabVIEW, LTSpice, Windchill, Zotero.